1 A student wishes to determine the Young modulus E of wood from the period of oscillation of a loaded wooden rule, as shown in Fig. 1.1.

For Examiner's Use

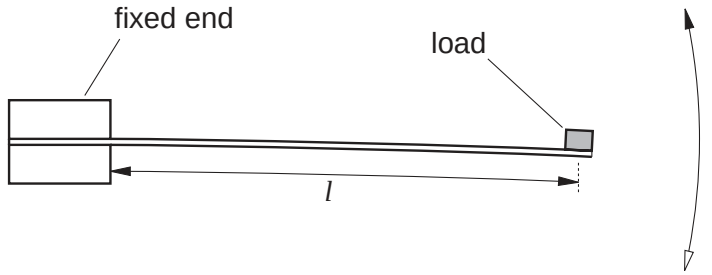

Fig. 1.1

An equation relating the period of oscillation T to the overhanging length l of the rule is

$$T^2 = \frac{kl^3}{E}.$$

The constant k is given by

$$k = \frac{16\pi^2 M}{wd^3}$$

where M is the mass of the load, w is the width of the rule and d is the thickness of the rule.

Design a laboratory experiment to determine the Young modulus of wood. You should draw a diagram showing the arrangement of your equipment. In your account, you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) how to analyse the data,

(e) how to determine E,

(f) the safety precautions to be taken.

[15]

© UCLES 2009

https://xtremepape.rs/